

Principal Component Analysis (PCA) of Client Satisfaction in Unemployment Benefit Services

Principal Component Analysis of Client Service Experience and Satisfaction

Dataset:	dole-survey (1).xlsx	Primary Software:	IBM SPSS Statistics
Analysis Files:	pcmF.xlsx, pcmF12.xlsx, pcmF1.xlsx	Analysis Method:	PCA with Varimax Rotation
Sample Size:	N = 388	Variance Explained:	52.11% (4 Components Retained)

Executive Summary

This research project examines client perceptions of service delivery using survey data collected from 388 respondents. The dataset contains demographic information, an overall satisfaction measure, 20 survey variables, and four standardized factor scores generated through Principal Component Analysis (PCA).

The primary objective of the analysis was to identify whether the 16 operational service-quality survey items could be reduced into a smaller number of meaningful underlying dimensions. PCA was selected as a data-reduction technique to identify the principal dimensions underlying client perceptions of service delivery.

The analysis demonstrated that the data were suitable for component extraction. The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was **0.712**, indicating an acceptable level of shared variance among the variables. Bartlett's Test of Sphericity was statistically significant, $\chi^2(120) = 680.630, p < .001$, confirming that the correlation matrix was not an identity matrix and that meaningful relationships existed among the survey items.

Using Kaiser's criterion of eigenvalues greater than 1.0, four components were retained. These four components collectively explained **52.11%** of the total variance in the 16-item dataset.

After Varimax rotation, the four dimensions were interpreted as:

- **Information & Administrative Transparency**
- **Caseworker Engagement & Effort**
- **Interpersonal Care & Respect**
- **Operational Reliability & Follow-Through**

The factor structure demonstrates that client perceptions of service quality are multidimensional rather than represented by a single generalized satisfaction construct. Clients evaluate the service experience through multiple interconnected dimensions, including the clarity of administrative processes, the quality of direct caseworker support, interpersonal treatment, and the organization's ability to fulfill commitments.

The analysis also showed that all 16 variables had acceptable communalities, ranging from **0.420 to 0.649**, indicating that each item shared meaningful variance with the retained component structure. Individual Measures of Sampling Adequacy (MSA) ranged from **0.582 to 0.773**, exceeding the commonly used minimum threshold of 0.50.

From a management perspective, the findings suggest that service improvement initiatives should not focus exclusively on overall satisfaction. Instead, organizations can use the four dimensions as a structured framework for monitoring and improving different aspects of the client experience.

1. Introduction

Public service organizations interact with clients across multiple stages of the service journey. These interactions may include administrative communication, application procedures, caseworker consultations, information provision, employment support, and follow-up activities.

Client satisfaction in such environments is therefore unlikely to be determined by a single factor. A client may be satisfied with the friendliness of staff but dissatisfied with the clarity of administrative procedures. Similarly, a client may receive useful job opportunities from a caseworker but still experience frustration if agreed actions are not completed.

The present analysis investigates the underlying structure of client perceptions using Principal Component Analysis. The purpose is to identify whether a large set of individual survey indicators can be summarized into a smaller number of interpretable service dimensions.

The resulting components can support:

- Research interpretation
- Service-quality measurement
- Management decision-making
- Dashboard development
- Performance monitoring
- Portfolio demonstration of statistical analysis skills

2. Research Objectives

The analysis was conducted with the following objectives:

- **Objective 1: Assess Factorability** — To determine whether the correlations among the survey items were sufficiently strong to justify dimension reduction through PCA.
- **Objective 2: Identify Underlying Dimensions** — To reduce 16 operational survey items into a smaller number of statistically meaningful components.
- **Objective 3: Determine the Number of Components** — To use eigenvalues and the scree plot to establish an appropriate number of components to retain.
- **Objective 4: Interpret the Component Structure** — To identify the main service-quality dimensions represented by the retained components.
- **Objective 5: Assess Model Adequacy** — To examine communalities, individual sampling adequacy, reproduced correlations, and residuals.
- **Objective 6: Translate Findings into Management Insights** — To convert the statistical findings into actionable recommendations for service improvement and performance monitoring.

3. Dataset Description

The primary dataset, `dole-survey (1).xlsx`, contains **388 observations** and **23 variables**.

Variable Group	Description
ssn	Respondent identifier
sex	Demographic variable
overall	Overall satisfaction/evaluation
v1-v20	Survey response variables
FAC1_1-FAC4_1	Standardized regression factor scores

The PCA was based on 16 operational survey items selected for the component analysis.

The survey items were measured primarily using an ordered response scale ranging from low agreement to high agreement, with a separate category representing no answer.

The analysis included item-level descriptive statistics, correlation analysis, PCA diagnostics, component extraction, Varimax rotation, communalities, reproduced correlations, residual analysis, and saved factor-score analysis.

4. Data Completeness and Quality

The dataset contains 388 total cases. The individual survey items demonstrated relatively high response completeness, with valid response rates generally ranging from approximately 95.9% to 98.5%. This indicates that item-level missingness was relatively limited.

However, an important distinction exists between the survey-item data and the saved factor scores. The four saved factor scores contain 244 non-missing observations, compared with 388 total respondents. This indicates that the factor-score generation procedure used a more restrictive complete-case approach, resulting in 144 missing factor-score observations.

Analytical Sample Size Summary:

- **Total dataset:** N = 388
- **PCA item-level analyses:** Variable-specific valid N values
- **Saved factor-score analyses:** N = 244 complete cases

This distinction is particularly important when comparing descriptive results with analyses involving FAC1_1 through FAC4_1.

5. Descriptive Analysis

The overall satisfaction variable had:

- **Valid N:** 381
- **Mean:** 4.276
- **Standard Deviation:** 1.412
- **Median:** 4 | **Minimum:** 1 | **Maximum:** 8

The item-level means indicate meaningful differences between service dimensions. The highest average scores were generally observed for items associated with interpersonal treatment. For example:

- v20 : Mean = 5.147
- v3 : Mean = 5.095
- v13 : Mean = 5.069
- v5 : Mean = 4.963

In contrast, several administrative and procedural items showed lower average scores. This suggests that clients may perceive interpersonal interactions more positively than administrative clarity and procedural transparency.

This distinction becomes particularly important in the subsequent PCA because the component structure confirms that different aspects of the service experience operate as distinct dimensions.

6. Correlation Analysis

The correlation matrix provided evidence of meaningful relationships among the 16 PCA variables. Several clusters of correlations were observed:

- **Interpersonal Service Cluster:** Items measuring privacy, friendliness, respect, and communication quality demonstrated positive relationships. Examples included Privacy and reception friendliness ($r = 0.439$), Privacy and feeling taken seriously ($r = 0.411$), Reception friendliness and feeling taken seriously ($r = 0.398$), and Reception friendliness and communication tone ($r = 0.402$). These relationships suggest a common underlying dimension related to interpersonal care and respectful treatment.
- **Caseworker Engagement Cluster:** Several caseworker-focused items were positively correlated. Examples included Motivation and fitting job opportunities ($r = 0.468$), Time spent with clients and fitting job opportunities ($r = 0.476$), and Time spent and interview preparation ($r = 0.367$). These relationships support the existence of an underlying dimension related to caseworker engagement and individualized effort.
- **Administrative Transparency Cluster:** Administrative information items also demonstrated meaningful relationships. Examples included Information accessibility and application progress ($r = 0.382$), Rights clarity and knowing who can answer questions ($r = 0.361$), and Rights clarity and application progress ($r = 0.359$). These relationships indicate a common dimension associated with administrative and information transparency.
- **Operational Reliability Cluster:** Items related to agreements, procedures, and promise fulfillment demonstrated positive relationships. For example, Agreements followed through and promises fulfilled ($r = 0.342$), and Agreements followed through and remaining procedures ($r = 0.291$). These relationships provide preliminary evidence of a reliability and follow-through dimension.

Overall, the correlation matrix contained sufficient relationships to justify formal factorability testing and PCA.

7. Assessment of Factorability

Before extracting components, the suitability of the dataset for PCA was assessed using:

- KMO Measure of Sampling Adequacy
- Bartlett's Test of Sphericity
- Anti-Image Correlation Matrix
- Correlation Matrix

8. KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure: The KMO value was **KMO = 0.712**. This indicates an acceptable level of sampling adequacy and suggests that the variables share enough common variance for PCA. A value above 0.70 is generally considered acceptable for exploratory dimension-reduction analysis. Therefore, the KMO result supports proceeding with PCA.

Bartlett's Test of Sphericity: The results were $\chi^2(120) = 680.630$, $p < .001$. The statistically significant result indicates that the null hypothesis that the correlation matrix is an identity matrix can be rejected. In practical terms, the survey variables are significantly correlated with one another.

Therefore, the combination of **KMO = 0.712** and **Bartlett's Test $p < .001$** provides evidence that the dataset is suitable for PCA.

9. Anti-Image Analysis

The Anti-Image Matrix provides an additional assessment of whether individual variables are suitable for inclusion in the PCA.

The individual Measures of Sampling Adequacy ranged from **0.582 to 0.773**. The lowest value was approximately 0.582, while the highest was approximately 0.773. All variables exceeded the commonly used minimum threshold of 0.50.

Therefore, no individual variable was identified as having insufficient sampling adequacy. This supports retaining all 16 variables in the final component solution.

10. Principal Component Extraction

Principal Component Analysis was used to reduce the 16 operational survey variables into a smaller set of dimensions. The extraction decision was based primarily on Kaiser's criterion: Retain components with eigenvalues greater than 1.0.

Component	Eigenvalue	Variance Explained
Component 1	2.675	16.72%
Component 2	2.308	14.43%
Component 3	1.857	11.61%
Component 4	1.496	9.35%

The four retained components collectively explained **52.11% of total variance**. Thus, the PCA reduced 16 original survey indicators into four major dimensions while retaining more than half of the total variance contained in the original variables.

11. Scree Plot Analysis

The scree plot provided additional visual evidence for retaining four components. The eigenvalue curve demonstrated a relatively sharp decline across the first four components. Beginning with Component 5, the curve became substantially flatter, with the fifth eigenvalue falling below 1 ($\lambda = 0.916$).

The combination of Kaiser's criterion, eigenvalue structure, scree plot pattern, and interpretability of the resulting components supports the selection of a four-component solution, which was retained for subsequent Varimax rotation and interpretation.

12. Communalities

Communalities indicate the proportion of each variable's variance accounted for by the retained components. The extracted communalities ranged from **0.420 to 0.649**.

Variable	Communality	Variable	Communality
v10	0.649	v12	0.514
v4	0.597	v16	0.508
v3	0.591	v14	0.484
v13	0.574	v15	0.470
v1	0.573	v9	0.459
v5	0.551	v8	0.450
v7	0.550	v2	0.422
v6	0.526	v11	0.420

All variables exceeded approximately 0.40, indicating acceptable representation within the four-component model. No variable was identified as having extremely low shared variance.

13. Rotated Component Structure

A Varimax orthogonal rotation was applied to improve interpretability and achieve a clearer component structure. The final four-component solution was interpreted using the saved factor-score relationships and the rotated component structure.

Factor 1: Information & Administrative Transparency

Saved Factor Score: FAC1_1

The following items showed strong relationships with FAC1_1: **v13 (0.703)**, **v16 (0.686)**, **v17 (0.667)**, **v9 (0.660)**, and **v2 (0.631)**.

This dimension represents the client's ability to understand and navigate the administrative environment. It includes clarity of unemployment benefit information, understanding of rights, accessibility of information, clarity regarding application progress, and knowing whom to contact for questions.

Interpretation: Factor 1 represents the degree to which the organization provides clients with understandable, accessible, and actionable administrative information. A low score on this factor may indicate that clients do not understand their rights, cannot easily locate information, are uncertain about application progress, do not know who to contact, or receive insufficiently clear benefit information. This factor is therefore closely related to transparency, accessibility, and administrative clarity.

Factor 2: Caseworker Engagement & Effort

Saved Factor Score: FAC2_1

The following items demonstrated strong relationships with FAC2_1: **v11 (0.790)**, **v7 (0.737)**, **v6 (0.716)**, and **v8 (0.669)**.

This dimension represents the quality and intensity of direct caseworker engagement. It includes providing suitable job opportunities, spending sufficient time with clients, motivating clients, and preparing carefully for interviews.

Interpretation: Factor 2 reflects the degree to which caseworkers provide individualized attention and active support. A high score indicates that clients perceive their caseworker as engaged, prepared, motivating, attentive, and helpful in identifying suitable employment opportunities. This factor is particularly relevant to workforce placement and personalized service delivery.

Factor 3: Interpersonal Care & Respect

Saved Factor Score: FAC3_1

The following variables demonstrated strong relationships with FAC3_1: **v3 (0.767)**, **v1 (0.750)**, **v20 (0.710)**, and **v5 (0.698)**.

This factor represents the human and interpersonal dimension of service delivery. It includes reception desk friendliness, protection of client privacy, feeling taken seriously, and appropriate communication tone.

Interpretation: Factor 3 reflects whether clients feel respected, valued, protected, and treated with dignity. The relatively high item means associated with this dimension suggest that interpersonal care is one of the stronger areas of perceived service quality in the dataset. However, maintaining this strength requires consistent attention to front-desk behavior, privacy protection, respectful communication, and client dignity.

Factor 4: Operational Reliability & Follow-Through

Saved Factor Score: FAC4_1

The following variables demonstrated strong relationships with FAC4_1: **v4 (0.769)**, **v14 (0.752)**, and **v12 (0.615)**.

This dimension represents the organization's ability to translate commitments and agreements into completed actions. It includes following through on agreements, keeping promises, and providing clear information about remaining procedures.

Interpretation: Factor 4 reflects operational trust and reliability. A high score indicates that clients perceive the organization as dependable and consistent in fulfilling commitments. A low score may indicate a gap between what clients are told will happen and what actually happens. This dimension is therefore particularly important for accountability, process consistency, trust, follow-up systems, and commitment tracking.

14. Summary of the Four-Factor Model

Factor	Dimension	Key Variables	Variance Contribution
FAC1_1	Information & Administrative Transparency	v2, v9, v13, v16, v17	16.72%
FAC2_1	Caseworker Engagement & Effort	v6, v7, v8, v11	14.43%
FAC3_1	Interpersonal Care & Respect	v1, v3, v5, v20	11.61%
FAC4_1	Operational Reliability & Follow-Through	v4, v12, v14	9.35%
Total	Four-component solution	16 items	52.11%

The four dimensions provide a practical framework for understanding client service experience.

15. Reproduced Correlations and Residuals

The reproduced correlation matrix was examined to determine how effectively the four-component solution reconstructed the original correlation structure. The residual matrix compares observed and reproduced correlations.

The analysis identified **38.0% of nonredundant residuals with absolute values greater than 0.05**. This result indicates that the four-component model reproduces a substantial proportion of the observed correlation structure. Under the criterion used in this analysis, where a model is considered acceptable if fewer than 50% of nonredundant residuals exceed an absolute value of 0.05, the four-component solution demonstrates an acceptable level of model representation. However, the residual result should be presented as supporting evidence of model adequacy, rather than as definitive proof of perfect model fit.

16. Component Transformation Matrix

The Component Transformation Matrix represents the mathematical transformation used to rotate the initial component solution into the final Varimax-rotated solution. The Varimax rotation seeks to improve interpretability by producing a simpler component structure, generally characterized by stronger primary loadings and weaker secondary loadings.

Because Varimax is an orthogonal rotation, the resulting components are mathematically constrained to remain uncorrelated in the rotated solution. The transformation matrix therefore provides technical documentation of how the initial component axes were rotated to achieve the final structure.

17. Component Plot in Rotated Space

The rotated component plot provides a visual representation of the spatial relationship among the survey variables. Variables associated with similar underlying constructs tend to appear closer together in the component space.

The plot visually supports the separation of items into groups associated with Administrative transparency, Caseworker engagement, Interpersonal care, and Operational reliability. The visual clustering provides an additional intuitive representation of the factor structure identified through the rotated component matrix.

18. Saved Factor Scores

SPSS generated four standardized regression factor scores: `FAC1_1`, `FAC2_1`, `FAC3_1`, and `FAC4_1`. Each factor score has approximately **Mean = 0** and **Standard deviation = 1**. The factor scores therefore represent standardized positions of respondents on the four underlying dimensions.

Factor Score	Minimum	Maximum
FAC1_1	-2.174	2.362
FAC2_1	-2.248	2.341
FAC3_1	-2.653	1.916
FAC4_1	-2.766	2.164

These factor scores can be used in downstream analyses, including regression analysis, group comparisons, demographic segmentation, predictive modeling, cluster analysis, and dashboard development. However, because only 244 cases have non-missing factor scores, any downstream analysis using these variables should explicitly report the reduced analytical sample.

19. Strategic Interpretation

The PCA demonstrates that client satisfaction is multidimensional. The four dimensions can be interpreted as a service delivery chain:

Information → Human Interaction → Individualized Support → Reliable Follow-Through

A client must first understand the system and know where to obtain information. They then interact with staff and caseworkers. The quality of direct support influences their experience. Finally, the organization must deliver on its commitments.

This means that improvements in one area may not fully compensate for weaknesses in another. For example, excellent front-desk friendliness cannot completely offset unclear application procedures. Similarly, a highly motivated caseworker may provide an excellent consultation, but client trust may still decline if promised follow-up actions are not completed.

20. Management Recommendations

Recommendation 1: Improve Information Transparency (Focus on Factor 1)

Recommended actions: Simplify benefit information, improve online information architecture, create clear application-process roadmaps, provide easy-to-understand explanations of client rights, clearly identify the appropriate contact person, and introduce centralized FAQ resources.

Expected Outcome: Clients should experience less uncertainty and require fewer repeated information requests.

Recommendation 2: Strengthen Caseworker Capacity (Focus on Factor 2)

Recommended actions: Review caseworker caseloads, allocate sufficient consultation time, improve interview preparation standards, provide better job-matching tools, and introduce structured motivational-support practices.

Expected Outcome: Clients should receive more individualized and relevant support.

Recommendation 3: Protect Interpersonal Service Quality (Focus on Factor 3)

Recommended actions: Continue front-desk customer-service training, maintain strict privacy practices, train employees in respectful communication, review written communication for tone and accessibility, and monitor client dignity and respect indicators.

Expected Outcome: The organization can maintain its stronger interpersonal performance while preventing service-quality deterioration.

Recommendation 4: Strengthen Operational Accountability (Focus on Factor 4)

Recommended actions: Introduce commitment-tracking systems, record agreed actions after consultations, send clients written summaries of next steps, use automated reminders for outstanding actions, and monitor completion of promised follow-ups.

Expected Outcome: Clients should experience greater consistency between commitments and actual service delivery.

21. Proposed Dashboard Framework

The four PCA dimensions can be translated into four executive-level service-quality KPIs.

Dimension	Suggested KPI	Management Focus
Information Transparency	Information Clarity Score	Digital information and process communication
Caseworker Engagement	Caseworker Support Score	Time, preparation, motivation, job matching
Interpersonal Care	Respect & Communication Score	Privacy, friendliness, dignity
Operational Reliability	Follow-Through Score	Commitments, promises, next steps

A dashboard could present these four dimensions as a Service Experience Scorecard. Recommended dashboard features include factor score trends, demographic filters, regional comparisons, office-level comparisons, time-based performance tracking, item-level drill-down, response counts, and missing-data indicators. If the dashboard uses weighted composite scores derived from factor loadings, these should be clearly labeled as analyst-created composite indices and distinguished from SPSS regression factor scores.

22. Proposed Implementation Roadmap

Phase 1: Quick Wins — Months 1–3

Focus: Information Transparency & Interpersonal Care

Actions: Simplify standard communication templates, improve online information navigation, clarify rights information, improve visibility of contact information, and reinforce privacy and respectful communication training.

Phase 2: Core Process Improvement — Months 4–6

Focus: Caseworker Engagement & Operational Reliability

Actions: Implement structured appointment preparation, introduce written meeting summaries, record commitments and agreed actions, improve job-matching support, and establish follow-up reminders.

Phase 3: Continuous Governance — Months 7–12

Focus: All four service dimensions

Actions: Launch recurring client pulse surveys, monitor factor scores quarterly, compare performance across offices

or regions, conduct regular caseload reviews, audit commitment fulfillment, and incorporate factor-based KPIs into management reviews.

23. Monitoring and Evaluation Framework

A proposed monitoring framework is:

Factor	Monitoring Indicator	Suggested Frequency
Information Transparency	Benefit information clarity and first-contact resolution	Monthly
Interpersonal Care	Respect and communication satisfaction	Quarterly
Caseworker Engagement	Suitable job-opportunity alignment	Quarterly
Operational Reliability	Commitment and follow-through completion	Monthly

These indicators should be treated as proposed management KPIs, not direct statistical outputs of the PCA. The PCA identifies the dimensions. The organization should then determine appropriate operational KPIs that measure performance within those dimensions.

24. Methodological Limitations

Several limitations should be acknowledged:

- **24.1 PCA versus Common Factor Analysis:** PCA is primarily a data-reduction technique rather than a latent-variable factor model. Therefore, the four components should be interpreted as dimensions summarizing the observed variables rather than definitive latent psychological constructs.
- **24.2 Variance Explained:** The four components explain 52.11% of total variance. This is a meaningful level of dimensional reduction, but approximately 47.89% of variance remains unexplained by the retained components. Other factors may influence client satisfaction, including benefit amount, employment status, economic conditions, regional differences, personal circumstances, waiting times, and previous service experiences.
- **24.3 Missing Data:** The saved factor scores are available for only 244 cases. This limits the usable sample size for downstream analyses involving factor scores.
- **24.4 Cross-Sectional Data:** The survey represents responses at a particular point in time. Therefore, causal relationships cannot be established.
- **24.5 Factor Interpretation:** The names assigned to the four components are analytical interpretations based on the variables that load most strongly on each component. The labels should therefore be treated as conceptual summaries rather than objectively measured organizational categories.
- **24.6 Generalizability:** The findings are based on this specific dataset and should not automatically be generalized to other agencies, populations, or geographic regions without additional validation.

25. Data Privacy and Portfolio Considerations

The dataset contains a variable identified as `ssn`, which is described as a respondent identifier. If this variable contains actual personally identifiable information, it should not be included in a publicly accessible GitHub repository or portfolio dataset.

For public documentation: remove direct identifiers, anonymize respondent IDs, do not publish raw personally identifiable information, use aggregated results where appropriate, and retain only variables necessary for demonstrating the analysis.

26. Final Conclusion

This Principal Component Analysis successfully reduced a set of 16 operational survey indicators into four interpretable service dimensions. The statistical diagnostics demonstrated that the dataset was appropriate for PCA: **KMO = 0.712**, **Bartlett's Test: $\chi^2(120) = 680.630$, $p < .001$** , Individual MSA: 0.582–0.773, and Communalities: 0.420–0.649. The four-component solution retained components with eigenvalues above 1.0 and explained 52.11% of total variance.

The final dimensions were interpreted as:

1. **Information & Administrative Transparency:** Clients' ability to understand benefits, rights, processes, information, and points of contact.
2. **Caseworker Engagement & Effort:** The quality of individualized support, including motivation, time, preparation, and job opportunities.
3. **Interpersonal Care & Respect:** The extent to which clients experience friendliness, privacy, dignity, respect, and appropriate communication.
4. **Operational Reliability & Follow-Through:** The organization's ability to keep promises, follow agreements, and clearly communicate remaining procedures.

The central conclusion is that client service experience is multidimensional. Effective service delivery requires more than positive interpersonal interactions. It requires a combination of clear information, respectful treatment, individualized support, and reliable follow-through.

The PCA framework provides a practical bridge between statistical analysis and organizational management. By transforming 16 individual survey indicators into four interpretable service dimensions, the organization can move from monitoring isolated survey questions toward a structured service-quality framework.

The resulting factor scores can also serve as inputs for future predictive analysis, segmentation, regression modeling, and business intelligence dashboards, provided that the reduced sample size associated with the saved factor scores is appropriately addressed.

Overall, the analysis demonstrates a complete applied data-analytics workflow, from raw survey data and descriptive analysis through correlation analysis, PCA diagnostics, component extraction, rotation, interpretation, and management recommendations.

Appendix A: Technical Analysis Summary

Parameter	Result
Dataset	dole-survey (1).xlsx
Total Cases	388
Total Variables	23
PCA Variables	16
Extraction Method	Principal Component Analysis
Rotation	Varimax with Kaiser Normalization
KMO	0.712
Bartlett's Test	$\chi^2 = 680.630$ (df = 120, p < .001)
Components Retained	4 (Kaiser Criterion: Eigenvalue > 1)
Variance Explained	52.11%
Communality Range	0.420–0.649
Individual MSA Range	0.582–0.773
Saved Factor Scores	FAC1_1–FAC4_1 (N = 244 complete cases)

Appendix B: Final Factor Mapping

Factor Score	Dimension	Variables
FAC1_1	Information & Administrative Transparency	v2, v9, v13, v16, v17
FAC2_1	Caseworker Engagement & Effort	v6, v7, v8, v11
FAC3_1	Interpersonal Care & Respect	v1, v3, v5, v20
FAC4_1	Operational Reliability & Follow-Through	v4, v12, v14

Appendix C: Factor Score Relationships

FAC1_1: v13: 0.703 | v16: 0.686 | v17: 0.667 | v9: 0.660 | v2: 0.631
FAC2_1: v11: 0.790 | v7: 0.737 | v6: 0.716 | v8: 0.669
FAC3_1: v3: 0.767 | v1: 0.750 | v20: 0.710 | v5: 0.698
FAC4_1: v4: 0.769 | v14: 0.752 | v12: 0.615

Appendix D: Suggested Portfolio Project Description

Project Title: Principal Component Analysis of Client Service Experience

Project Summary: Conducted an end-to-end Principal Component Analysis using IBM SPSS Statistics on 388 client survey responses to identify the underlying dimensions of service experience. Assessed factorability using KMO, Bartlett's Test, and Anti-Image diagnostics; extracted and rotated four components using PCA and Varimax rotation; evaluated eigenvalues, explained variance, communalities, reproduced correlations, and residuals; and translated statistical findings into actionable service-quality recommendations and a proposed management dashboard framework.

Key Result: Reduced 16 operational service-quality indicators into four interpretable dimensions—Information & Administrative Transparency, Caseworker Engagement & Effort, Interpersonal Care & Respect, and Operational Reliability & Follow-Through—collectively explaining 52.11% of total variance.

Tools: IBM SPSS Statistics, Microsoft Excel, PCA, Correlation Analysis, Factor Analysis Diagnostics, Data Visualization, Statistical Reporting.